

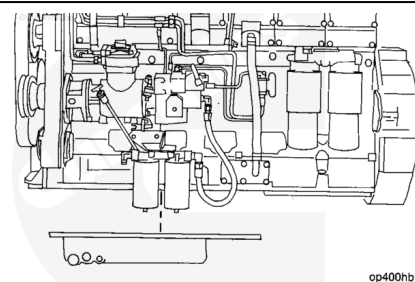
Trainings ▾

General Information

Lubricating oil pan sumps are available in various capacities. Check the part number and refer to Procedure 018-017 (</qs3/pubsys2/xml/en/procedures/20/20-018-017.html>) for the correct high and low capacities. When the rear gear train option is specified, add 7.6 liters [2 gallons] to the sump capacity.

Service dipsticks are available in two pan types. The first is the locking type. The second is the bottle stopper type that does **not** lock. When converting from one style to the other, change both the dipstick and the oil gauge tube.

Service dipsticks are supplied by length. The dipsticks do **not** have the high and low marks indicated. Do **not** calibrate the dipstick until the engine is installed in the application that it is to be used on.



Preparatory Steps

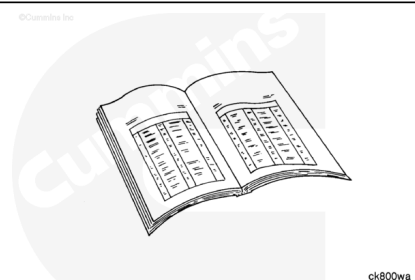
All Applications Except Rail

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

- Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>).

Rail Applications

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⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>).
- Drain the coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).
- Remove the lubricating oil filter (remote mounted). Refer to Procedure 007-017

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(/qs3/pubsys2/xml/en/procedures/20/20-007-017.html).

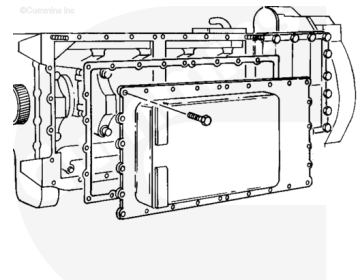
- Remove the fuel filter (remote mounted). Refer to Procedure 006-011 (/qs3/pubsys2/xml/en/procedures/20/20-006-011-tr.html).
- Remove the coolant filter (remote mounted). Refer to Procedure 008-008 (/qs3/pubsys2/xml/en/procedures/20/20-008-008.html).

Remove

All Applications Except Rail

⚠ WARNING ⚠

The component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.



Remove the 28 capscrews.

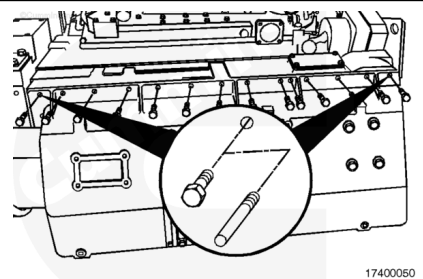
Remove the lubricating oil pan.

Remove and discard the gasket.

Rail Applications

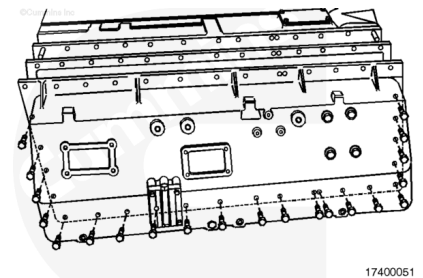
Remove 15 capscrews on the top of the lubricating oil pan.

Install two 3/8- 16 guide studs in the capscrew holes.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.



Remove the remaining 23 capscrews.

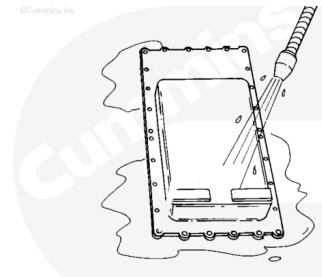
Remove the lubricating oil pan.

Remove and discard the gasket.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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⚠ CAUTION ⚠

Use solvent that will not harm aluminum.

Clean the lubricating oil pan with solvent.

Inspect for cracks or other damage.

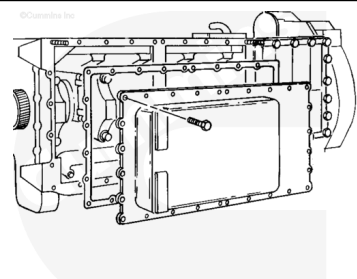
If the lubricating oil pan is cracked or otherwise damaged it **must** be repaired or replaced.

Install

All Applications Except Rail

⚠ WARNING ⚠

The component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.



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⚠ CAUTION ⚠

Gasket cement will prevent the gasket from sealing properly.

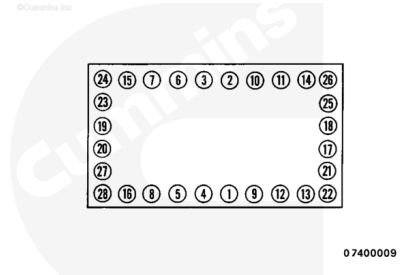
Use a contact adhesive such as 3M Spray 77 or 3M 4693 to hold the gasket in position.

Install the gaskets and the lubricating oil pan.

Install the capscrews.

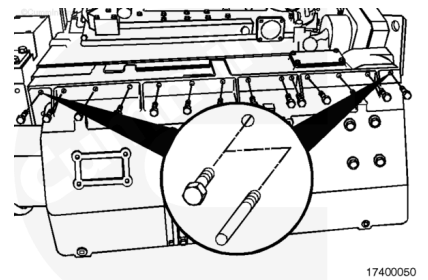
Tighten the capscrews in sequence.

Torque Value: 45 n•m [33 ft-lb]



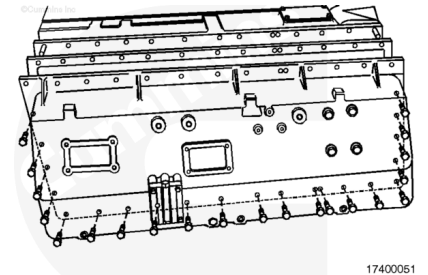
Rail Applications

Install two 3/8 - 16 guide studs in the top row of capscrew holes.



⚠ WARNING ⚠

The component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.



⚠ CAUTION ⚠

Gasket cement will prevent the gasket from sealing properly.

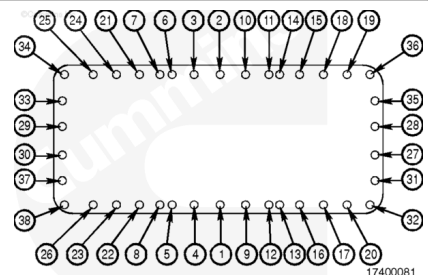
Use a contact adhesive such as 3M Spray 77 or 3M 4693 to hold the gasket in position.

Install the gasket and lubricating oil pan.

Install the 38 capscrews.

Tighten the capscrews in the sequence shown.

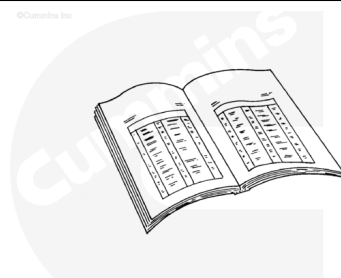
Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

All Applications Except Rail

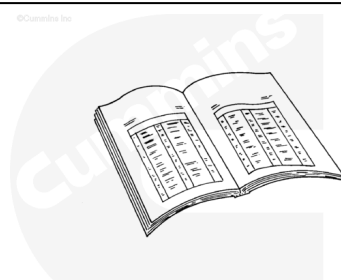
- Fill the engine with lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>).
- Operate the engine and check for leaks.



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Rail Applications

- Install the coolant filter (remote mounted). Refer to Procedure 008-008 (</qs3/pubsys2/xml/en/procedures/20/20-008-008.html>).
- Install the fuel filter (remote mounted). Refer to Procedure 006-011 (</qs3/pubsys2/xml/en/procedures/20/20-006-011-tr.html>).
- Install the lubricating oil filter (remote mounted). Refer to Procedure 007-017 (</qs3/pubsys2/xml/en/procedures/20/20-007-017.html>).
- Fill cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).
- Fill the engine with lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>).
- Operate the engine and check for leaks.



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